

Traffic Monitoring System

<https://github.com/aiyshawry/Traffic-Monitoring-System>

PHP

MySQL

JavaScript

HTML

CSS

XAMPP

OVERVIEW

A PHP and SQL-based web application that supports traffic controllers with real-time violator tracking, periodic traffic status notifications, daily reports, and transport department recommendations for infrastructure improvements.

IMPACT

Developed a multi-role Traffic Monitoring System web application with separate portals for admins, officers, and citizens, enabling automated violator tracking, fine management, road signalling status, and daily traffic reporting.

TECHNICAL WALKTHROUGH

Traffic Monitoring System (TMS) Date: March 8, 2026 Built with PHP, MySQL, Apache (MAMP) | Bootstrap 4 Frontend

Project Overview

The Traffic Monitoring System (TMS) is a web-based application designed to streamline traffic management for both authorities (Officers) and the public (Citizens). It provides tools for monitoring road conditions, managing traffic signal timings, and handling traffic violation fines - including online payment. The system is built using PHP for server-side logic, MySQL as the relational database, Apache as the web server (via MAMP), and Bootstrap 4 for a responsive frontend UI.

Technology Stack

Layer Technology / Library Backend Language PHP 8.3 (server-side scripting) Database MySQL 8 (via MySQLi extension) Web Server Apache (MAMP on macOS) Frontend Framework Bootstrap 4 JavaScript Libraries jQuery 2/3, Owl Carousel, ScrollReveal CSS Libraries Font Awesome 4.7, Animate.css, Perfect Scrollbar UI Utilities Select2, Tilt.js, Lightbox Session Management PHP Native Sessions

User Roles

The system has three distinct user roles, each identified by a file prefix convention: Prefix Role Access Level
a Admin / Public Landing page, login/register chooser. No authentication required.
c Citizen Registered vehicle owners. Can view own fines, pay fines, view signals and roads.
o Officer Traffic authority personnel. Can manage fines, edit signal timings, view road data.

File Structure & Purpose

4.1 Shared / Core Files File Purpose `aconnection.php` MySQL DB connection - included by all PHP pages. Holds host, user, pass, dbname. `logout.php` Destroys the PHP session and redirects the user back to `ahome.php`. `database.sql` Full MySQL schema - creates all 5 tables and inserts sample road/signal data.

4.2 Public / Admin Pages (prefix: a) File Purpose `ahome.php` Public homepage. Slider, About, Features, Team, Contact sections. `alioption.php` Login chooser page - two buttons: Citizen Login or Officer Login. `asuooption.php` Register chooser page - two buttons: Citizen Signup or Officer Signup. `alogout.php` Legacy logout file kept for compatibility.

4.3 Citizen Pages (prefix: c) File Purpose `csignup.php` Registration form. Inserts into 'citizen' table AND auto-creates a fine record. `clogin.php` Login form. Validates against 'citizen' table, sets `$_SESSION['email']`. `cfunctions.php` `check_login()` function - validates session, redirects to `clogin.php` if absent. `chome.php` Citizen dashboard with slider linking to all citizen features. `cfine.php` Shows own fine records (amount, status). Pay Now button if unpaid. `cpayment.php` Card payment UI. On submit sets `fine.paid = 1` in database. `csignalling.php` Read-only view of signal timings (road name, `time_from`, `time_to`). `croad.php` Read-only view of all roads (station name, road name).

4.4 Officer Pages (prefix: o) File Purpose `osignup.php` Registration form for officers. Stores username, email, OID, password. `ologin.php` Login form. Validates against 'officer' table, sets `$_SESSION['email']`. `ofunctions.php` `check_login()` function - validates session, redirects to `ologin.php` if absent. `ohome.php` Officer dashboard with slider linking to all officer features. `ofine.php` Lists ALL fines in the system with an Edit button per row. `ozfupdate.php` Edit form to update a specific fine amount by fine ID. `osignalling.php` Lists all signal timings with an Edit button per row. `osupdate.php` Edit form to update `time_from` and `time_to` for a signal entry. `oroad.php` View all roads - Road ID, Station Name, Road Name.

Database Schema

Database name: 'project' | Character set: utf8mb4 | Engine: InnoDB

Table: citizen - Stores registered citizens

Column	Type	Constraints
id	INT	AUTO_INCREMENT, PRIMARY KEY
username	VARCHAR(100)	NOT NULL
email	VARCHAR(100)	NOT NULL, UNIQUE
Phone	VARCHAR(15)	
VID	VARCHAR(20)	
Vehicle ID	pass	VARCHAR(255) NOT NULL (plain text - upgrade to hashed in production)

Table: officer - Stores traffic officers

Column	Type	Constraints
id	INT	AUTO_INCREMENT, PRIMARY KEY
username	VARCHAR(100)	NOT NULL
email	VARCHAR(100)	NOT NULL, UNIQUE
OID	VARCHAR(20)	Officer ID
pass	VARCHAR(255)	NOT NULL

Table: fine - Traffic violation fines

Column	Type	Constraints
id	INT	AUTO_INCREMENT, PRIMARY KEY
username	VARCHAR(100)	Links to citizen.username
VID	VARCHAR(20)	Vehicle ID
amount	DECIMAL(10,2)	Default 0.00
paid	TINYINT	0 = Pending, 1 = Paid

Table: road - Road information

Column	Type	Constraints
id	INT	AUTO_INCREMENT, PRIMARY KEY
station_name	VARCHAR(100)	
road_name	VARCHAR(100)	

Table: signalling - Traffic signal timings per road

Column	Type	Constraints
id	INT	AUTO_INCREMENT, PRIMARY KEY
road_name	VARCHAR(100)	
time_from	VARCHAR(20)	e.g. 06:00
time_to	VARCHAR(20)	e.g. 10:00

Complete Application Flow

6.1 Citizen Flow

Step Page Action

Step 1 Visit `ahome.php` Public homepage. Click 'Register Now'.

Step 2 `asuooption.php` Choose 'CITIZEN' to go to `csignup.php`.

Step 3 `csignup.php` (POST) Fill name, email, phone, vehicle ID, password. Two DB inserts: citizen table + fine table (`amount=0`).

Step 4 `clogin.php` (POST) Enter email + password. Session set on success. Redirect to `chome.php`.

Step 5 `chome.php` Dashboard. Links to Fine Payment, Signal Timings, Google Maps, Road Info.

Step 6 `cfine.php` View own fine. If `amount > 0` and unpaid, 'Pay Now' link appears.

Step 7 `cpayment.php` Enter card details. On submit: `fine.paid = 1` in DB. Redirected back with success message.

Step 8 `logout.php` Session destroyed. Redirect to `ahome.php`.

6.2 Officer Flow

Step Page Action

Step 1 `ahome.php` Click 'Register Now'.

Step 2 `asuooption.php` Choose 'OFFICER' to go to `osignup.php`.

Step 3 `osignup.php` (POST) Fill name, email, Officer ID, password. Insert into officer table.

Step 4 `ologin.php` (POST) Login. Session set. Redirect to `ohome.php`.

Step 5 ohome.php Dashboard. Links to Fine Allotment, Signal Timings, Google Maps, Road Info.

Step 6 ofine.php View ALL citizen fines. Each row has an Edit link.

Step 7 ozfupdate.php Edit fine amount for a specific citizen. Updates fine.amount in DB.

Step 8 osignalling.php View all signal timings. Each row has an Edit link.

Step 9 osupdate.php Update time_from and time_to for a road signal.

Step 10 oroad.php View all road records (read-only).

Step 11 logout.php Session destroyed. Redirect to ahome.php.

Authentication & Session Management

Both Citizens and Officers use PHP's native session mechanism: Login: `$_SESSION['email'] = $user_data['email'];` (set on successful login) Guard: Every protected page includes `check_login($con)` from the role's functions file.

If the session is not set, the user is immediately redirected to the login page.

Logout: `unset($_SESSION['email']); header('Location: ahome.php');` Two separate `check_login()` functions exist: `cfunctions.php` - queries the 'citizen' table, redirects to `clogin.php` if unauthenticated. `ofunctions.php` - queries the 'officer' table, redirects to `ologin.php` if unauthenticated.

Setup & Deployment Guide

Requirements MAMP (or XAMPP) with Apache + MySQL PHP 8.x Browser (Safari, Chrome, Firefox)

Installation Steps

Copy project folder to /Applications/MAMP/htdocs/Traffic-Monitoring-System/

Open MAMP and click 'Start Servers' (both Apache and MySQL must be green).

Open phpMyAdmin at <http://localhost:8888/phpMyAdmin>

Create a new database named 'project'.

Select 'project' database -> Import tab -> choose database.sql -> click Go.

Open the site at <http://localhost:8888/Traffic-Monitoring-System/ahome.php>

Database Credentials (`aconnection.php`) Parameter Value Host localhost Username root Password root (MAMP default) Database project

Security Notes & Known Limitations

Issue Detail SQL Injection Queries use raw string interpolation. Use prepared statements (PDO/MySQLi) in production.

Plain-text passwords Passwords stored as plain text. Use `password_hash()` and `password_verify()` in production.

No admin role There is no separate admin login. `ahome.php` is publicly accessible.

No input sanitization User input is not sanitised before DB insertion. Add `htmlspecialchars/strip_tags`.

Payment is simulated `cpayment.php` marks fine as paid but does not connect to a real payment gateway.

No CSRF protection Forms have no CSRF tokens. Add tokens to all POST forms in production.

Frontend Assets & Directory Layout

Directory Contents `assets/css/` Bootstrap, Font Awesome, Owl Carousel, Lightbox, main template CSS `assets/js/` jQuery, Bootstrap JS, Owl Carousel, ScrollReveal, Lightbox, custom.js `assets/images/` Slider images (a.jpg, b.jpg, c.jpg), icons, team photos `vendor/bootstrap/` Bootstrap 4 CSS + JS `vendor/animate/` Animate.css for scroll animations `vendor/select2/` Select2 dropdown library `vendor/perfect-scrollbar/` Smooth scrollbar for tables `vendor/tilt/` Tilt.js - 3D tilt effect on login images `vendor/css-hamburgers/` Hamburger menu animation CSS `css/` util.css + main.css - login page layout styles font-awesome-4.7.0/ Icon font library `images/icons/` Favicon and SVG illustrations (citizen.svg, officer.svg, traffic.svg)

KEY DETAILS

- Admin portal manages user accounts, officer assignments, road data, and signalling configurations.
- Officer portal logs violators, records fines, and updates real-time road and signal status.
- Citizen portal supports account registration, violation history lookup, and online fine payment.
- Automated daily reports summarize traffic violations and flag roads needing infrastructure attention.
- Built with PHP, MySQL (XAMPP), JavaScript, and CSS with a responsive multi-role front-end.